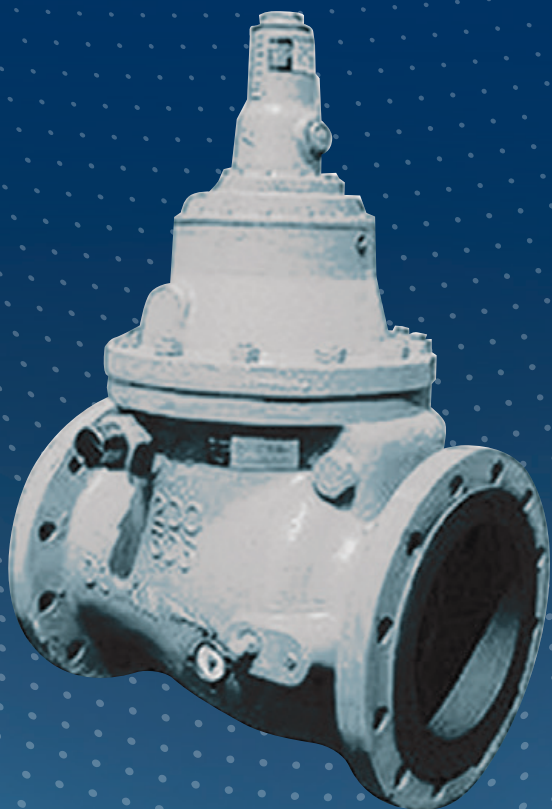


Safety Cut-Off Valve HON 303 MK5



PRODUCT INFORMATION

**Serving the Gas Industry
Worldwide**

Honeywell

SAFETY CUT-OFF VALVE HON 303 MK5

Introduction, Application, Features, Technical Data

Introduction

- The HON 303 is a safety cut-off valve of single in line orifice design, overpressure operated, with external control line connection, manual reset and pressure balancing bypass. Completely self-acting and requires no separate power source to close the valve.
- Use with confidence on natural and manufactured gases of non-aggressive nature, including Nitrogen, Carbon Dioxide, Propane, Butane etc.

Application

Designed to safeguard gas distribution systems, normally installed upstream of the pressure regulator with the external control line piped to the downstream of the regulator, closing automatically in the event of an overpressure condition. After closure, the valve must be manually reset to the open position after normal pressure conditions have been restored by use of the manual bypass valve.

Features

- Precise and dependable operation
- Fully enclosed valve door position indicator - colour coded to show valve status (Open/Closed).
- Specially designed spindle and seal between trigger mechanism and diaphragm actuator practically eliminates effect of inlet pressure variation on trip set-point.
- Pressure equalising valve
- Manual Reset

Size Range

DN 250 Flanged

DN 300 Flanged

Connections to PN16:BS EN1092-2:1997

(can be drilled to other standards on request)

SPRING SELECTION			
Model	Spring		Trip Range in mbarg
	Number	Colour	
303 LP	879	Grey	37 to 170
	880	Brown	160 to 480
303 MP	756	Silver	420 to 550
	757	Orange	540 to 760
	758	Yellow	750 to 970
	759	Black	950 to 1250
	760	White	1230 to 1600
	761	Lt. Green	1580 to 2000
	762	Pink	1970 to 2630
	763	Lt. Blue	2600 to 3460
	764	Gold	3430 to 4500

OPTIONS	
OPTION: 1	
Can be fitted with Pepperl & Fuchs Kontex proximity sensor system, to initiate 'Valve Open' or 'Valve Closed' indicator on remote panel.	
OPTION: 2	
Can be fitted with Encapsulated Tilt Switch assembly, to initiate 'Valve Open' or 'Valve Closed' indicator on remote panel.	

Installation

For installation in a horizontal pipeline with spring housing pointing vertically upwards and aligned with the gas flow direction, as indicated by the arrow on the valve body.

SERVICE CONDITIONS	
Maximum Inlet Pressure	7 barg
Pressure Range: 303LP 303MP	0.037 to 0.48 barg 0.42 to 4.5 barg
Temperature Range:	-20°C to +60°C

MATERIALS OF CONSTRUCTION	
Component	Material
Body, Top Cover, Spring Housing & Top Cap	Ductile Iron to: BSEN 1563 Grade: EN-GJS-400-18
Operating levers, Link plates, Valve door Diaphragm plates	Carbon Steel (Zinc Plated)
Torsion & compressor springs	Carbon Spring Steel (Lanolised)
Trigger block, Spindle bushes and bypass valve	Brass: BS249
Diaphragm	Nylon Reinforced Pentane Resistant Nitrile
Door, Operating spindles and bushes	Stainless Steel: BS970 - 416S21
Spring Adjuster: LP MP	Aluminium C800 Cast Iron: BS1452 220

SAFETY CUT-OFF VALVE HON 303 MK5

Pressure Loss, Materials of Construction

Pressure Loss

The HON 303 is designed for minimal pressure loss (ΔP).

$$P_i - P_o = \left(\frac{Q}{K} \right)^2 \times \frac{1}{P_o}$$

Where:

Q = Valve flow M³/Hr (SG 0.6)

P_i = Inlet Pressure bar (abs)

P_o = Outlet Pressure bar (abs)

K = Flow Constant

(abs) = Absolute pressure 1.01325

When sizing valves in accordance with GIS/V9-1, the maximum recommended gas velocity through the safety cut-off valve, should not exceed 80 m/sec.

Consideration given to higher gas velocities on request - please contact Honeywell for details.

FLOW CONSTANT

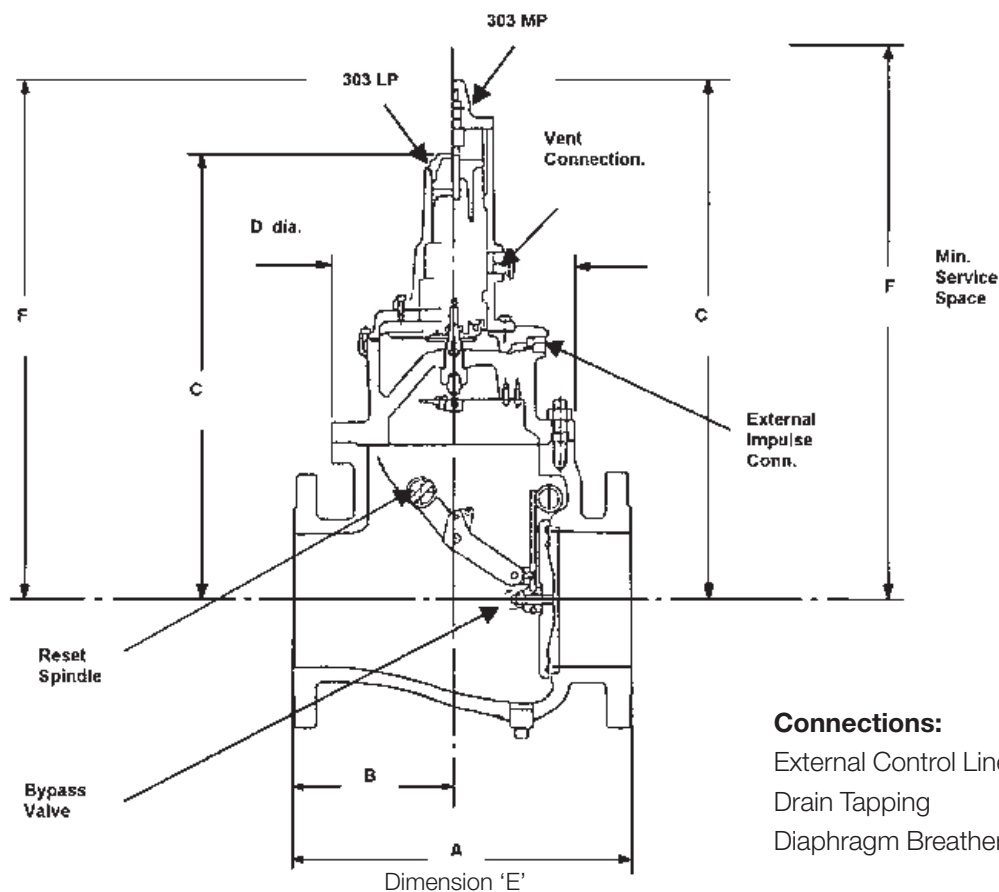
Valve Size	Flow Constant
DN 250	102675
DN 300	176600

Conversion Factor

The pressure loss calculation is given in terms of natural gas (SG 0.6). For all other gases multiply by the following correction factor

$$\sqrt{\frac{0.6}{\text{SG of Gas handled}}}$$

Conversion to ft³/hr - multiply by 35.3



Centre line valve to max. Extent bypass valve

Connections:

External Control Line	BS21 Rc 1/4
Drain Tapping	BS21 Rp 1
Diaphragm Breather Vent	BS21 Rc 3/4

DIMENSIONS & WEIGHTS

Size	A	B	C		D	E	F		Weight Kg's	
			LP	MP			LP	MP	LP	MP
DN 250	533	241	639	721	406	259	856	897	158	167
DN 300	610	279	698	779	489	292	915	956	214	228

All Dimensions in mm

For More Information

To learn more about Honeywell's
Advanced Gas Solutions, visit
www.honeywellprocess.com or contact
your Honeywell account manager.

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The Honeywell logo, consisting of the word "Honeywell" in a bold, red, sans-serif font.

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